

Advanced Memory Circuits and Systems

Assignment - 1

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GitHub link for all files: [Assignment 1 Link](#)

A ngspice: Read a 6T SRAM Cell and Measure the Margin Simulation Setup

The nominal transistor dimensions are

$$W_P = 0.15 \mu\text{m}, \quad W_N = 0.20 \mu\text{m}, \quad W_{\text{access}} = 0.16 \mu\text{m},$$

with

$$L = 0.045 \mu\text{m}.$$

The bitline capacitances are

$$C_{BL} = C_{BLB} = 180 \text{ fF},$$

and both bitlines are initially charged to

$$V_{BL} = V_{BLB} = V_{DD}.$$

A transient simulation is performed for 4 ns with a maximum timestep of 5 ps. The read differential is measured as

$$\Delta V(BL, BLB) = V(BLB) - V(BL)$$

at

$$t = 2 \text{ ns},$$

while the maximum read disturbance is obtained from

$$V_{Q,\text{max}} = \max_{1 \text{ ns} \leq t \leq 4 \text{ ns}} V(Q).$$

A.1 Bitline Differential Voltage

For the nominal configuration,

$$V_{DD} = 1.1 \text{ V}, \quad W_{\text{access}} = 0.16 \mu\text{m},$$

the measured differential voltage at $t = 2 \text{ ns}$ is

$$\Delta V(BL, BLB) = 0.689041 \text{ V} = 689.041 \text{ mV}$$

The same simulation gives

$$V_{Q,\max} = 0.198812 \text{ V}.$$

Parameter	Value
V_{DD}	1.1 V
W_{access}	0.16 μm
Measurement time	2 ns
$\Delta V(BL, BLB)$	689.041 mV
$V_{Q,\max}$	198.812 mV

A.1.1 Comparison with the 69 mV Lecture Value

The 69 mV value from Lecture 1 corresponds to the DRAM charge-sharing example rather than to a 6T SRAM read.

In that example,

$$V_{DD} = 1.1 \text{ V}, \quad C_S = 25 \text{ fF}, \quad C_{BL} = 175 \text{ fF},$$

and the bitline is initially precharged to $V_{DD}/2$. The bitline swing is calculated as

$$\Delta V = \frac{V_{DD}}{2} \frac{C_S}{C_S + C_{BL}} = 0.55 \frac{25}{25 + 175} \approx 69 \text{ mV}.$$

This is a DRAM charge-sharing mechanism. The storage capacitor and the bitline capacitor redistribute charge, resulting in a relatively small voltage perturbation.

The SRAM experiment is fundamentally different. Both bitlines are initially at V_{DD} , and once the wordline is asserted, the bitline associated with the stored-zero node is actively discharged through the access transistor and the cell pull-down transistor. Consequently, the differential can become much larger during the fixed 2 ns read interval.

Therefore,

$$\Delta V_{\text{SRAM}} = 689.041 \text{ mV}$$

and

$$\Delta V_{\text{DRAM}} = 69 \text{ mV}$$

are results from different memory cells and different sensing mechanisms.

The waveform shows that Q initially remains near zero while QB remains close to V_{DD} . After the wordline is asserted, the read operation causes a disturbance on the storage node and creates a differential between the two bitlines.

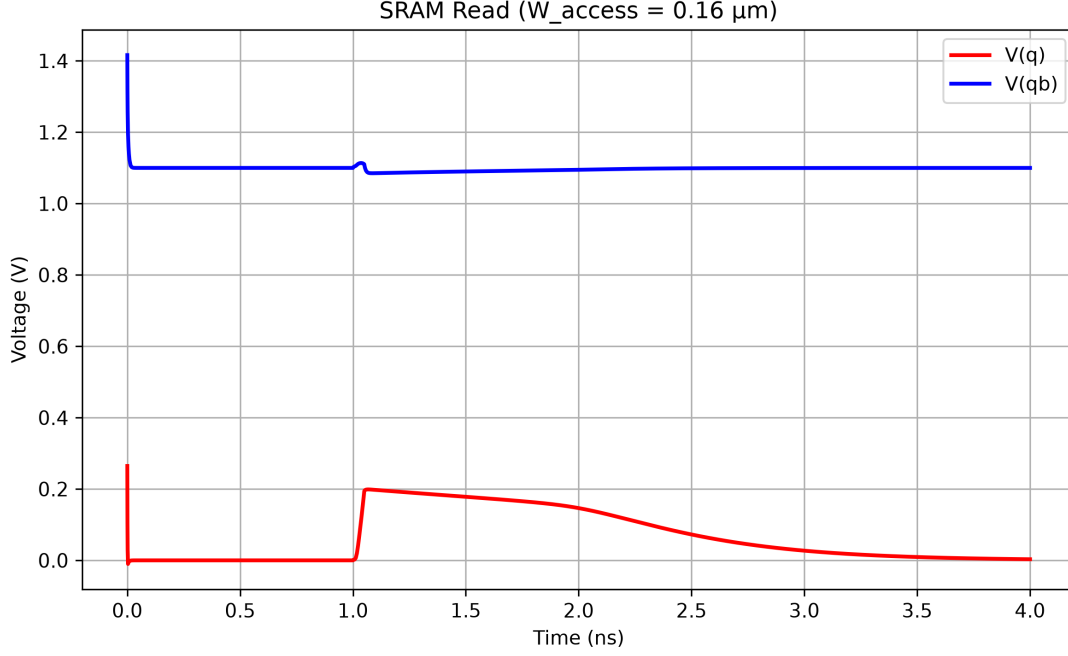


Figure 1: Transient response of the nominal 6T SRAM read operation for $W_{\text{access}} = 0.16 \mu\text{m}$.

A.2 Read Disturb and Access-Transistor Sizing

During a read, the stored-low node Q is connected to a precharged bitline through the access transistor. This creates a competing path against the cell pull-down transistor that attempts to keep Q at ground. Therefore, a conventional 6T SRAM read causes Q to rise above zero. This voltage rise is the read disturb. For

$$W_{\text{access}} = 0.16 \mu\text{m},$$

the maximum storage-node voltage is

$$V_{Q,\text{max}} = 198.812 \text{ mV}.$$

The access-transistor width was then increased to

$$W_{\text{access}} = 0.24 \mu\text{m}.$$

The resulting simulation gives

$$\Delta V(BL, BLB) = 815.023 \text{ mV}$$

$$V_{Q,\text{max}} = 276.333 \text{ mV}.$$

Parameter	$W_{\text{access}} = 0.16 \mu\text{m}$	$W_{\text{access}} = 0.24 \mu\text{m}$	%change
$\Delta V(BL, BLB)$	689.041 mV	815.023 mV	18.3%
$V_{Q,\text{max}}$	198.812 mV	276.333 mV	39%

The result demonstrates a fundamental SRAM sizing trade-off. A larger access transistor strengthens the connection between the cell and the bitline, which increases the sensing signal,

$$W_{\text{access}} \uparrow \Rightarrow \Delta V_{BL} \uparrow.$$

However, the same stronger access path increases the disturbance of the stored zero,

$$W_{\text{access}} \uparrow \Rightarrow V_{Q,\text{max}} \uparrow.$$

Thus, the access transistor must be sized by considering both sensing strength and read stability.

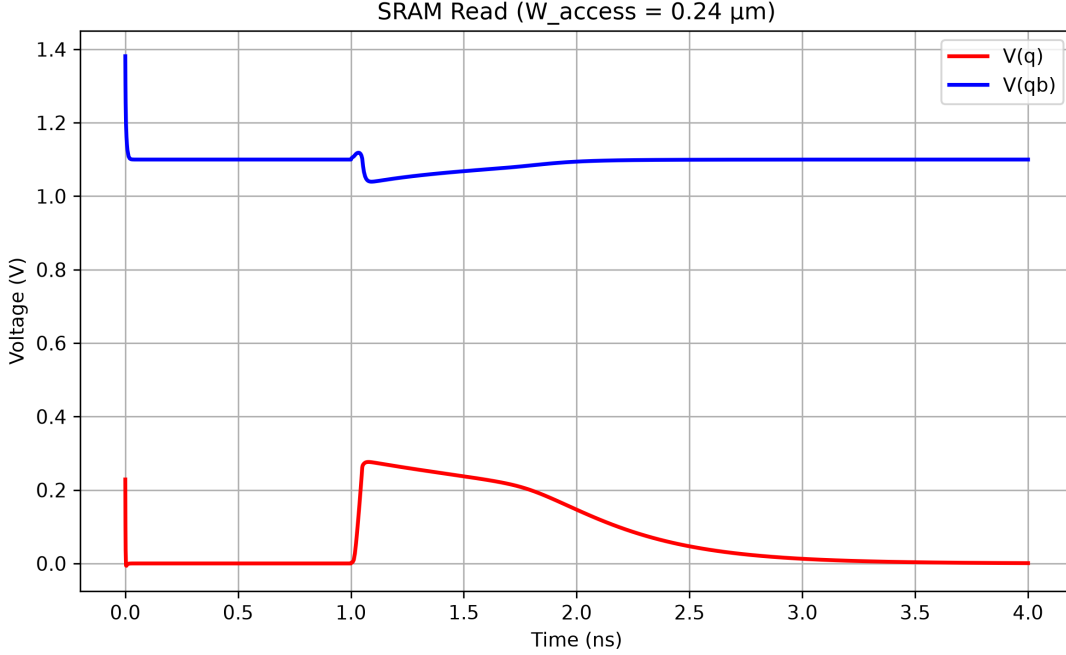


Figure 2: SRAM read transient for $W_{\text{access}} = 0.24 \mu\text{m}$.

The larger access transistor therefore produces a stronger read signal but also pushes the cell closer to read failure by increasing the disturbance of the stored state.

A.3 Read Margin versus V_{DD}

The supply voltage was swept from 1.1 V downwards in 50 mV steps. The bitline differential was measured at $t = 2$ ns for every supply voltage.

The read differential decreases continuously as the supply voltage is reduced. This is expected because the available voltage for discharging the bitline decreases and the transistor drive strength also becomes weaker.

The sense-amplifier offset is

$$V_{\text{SA,offset}} = 25 \text{ mV}.$$

At the lowest voltage explicitly required by the assignment, $V_{DD} = 0.6$ V, measured differential is

$$\Delta V = 190.507 \text{ mV},$$

which is still much larger than the 25 mV offset.
Therefore,

the sense-amplifier threshold is not crossed within the required 0.6–1.1 V range.

The sweep was extended below 0.6 V. The threshold crossing occurs between 0.40 and 0.45 V, since

$$\Delta V(0.45 \text{ V}) = 45.604 \text{ mV} > 25 \text{ mV}$$

while

$$\Delta V(0.40 \text{ V}) = 23.514 \text{ mV} < 25 \text{ mV}.$$

Using linear interpolation,

$$V_{DD,\text{crit}} = 0.40 + \frac{25 - 23.514}{45.604 - 23.514}(0.45 - 0.40),$$

which gives

$$V_{DD,\text{crit}} \approx 0.403 \text{ V}.$$

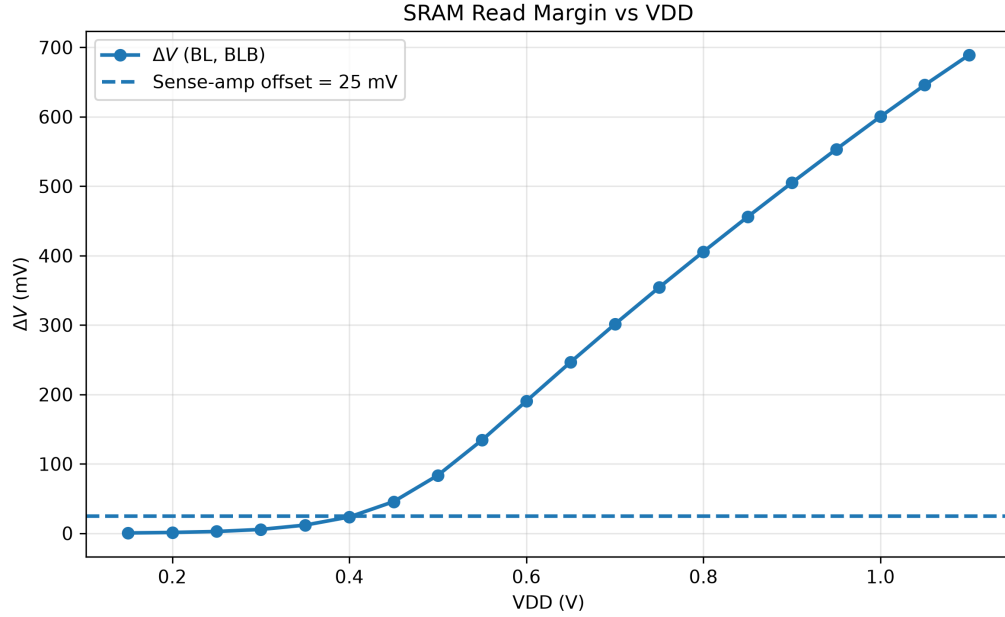


Figure 3: SRAM read differential as a function of V_{DD} . The dashed horizontal line is 25 mV.

A.4 Effect of Temperature

The nominal read operation was repeated at

$$T = 85^\circ\text{C}.$$

The measured differential is

$$\Delta V_{85^{\circ}\text{C}} = 0.582287 \text{ V} = 582.287 \text{ mV}.$$

The corresponding maximum voltage on the storage node is

$$V_{Q,\text{max}} = 218.750 \text{ mV}.$$

Compared with the nominal differential,

$$\Delta V_{\text{nominal}} = 689.041 \text{ mV},$$

The percentage reduction is

$$\frac{689.041 - 582.287}{689.041} \times 100 \approx 15.49\%.$$

Parameter	Nominal	85°C
$\Delta V(BL, BLB)$	689.041 mV	582.287 mV
$V_{Q,\text{max}}$	198.812 mV	218.750 mV

The sense-amplifier offset is treated as the fixed 25 mV reference specified in the assignment. Hence, the quantity that changes substantially in the temperature experiment is the SRAM-generated differential voltage.

At 85°C,

$$\frac{\Delta V}{V_{\text{SA,offset}}} = \frac{582.287}{25} \approx 23.3.$$

Therefore, even at elevated temperature, the available read differential is much larger than the assumed sense-amplifier offset.

The maximum storage-node voltage also increases from 198.812 mV to 218.750 mV, indicating a somewhat larger read disturbance at elevated temperature.

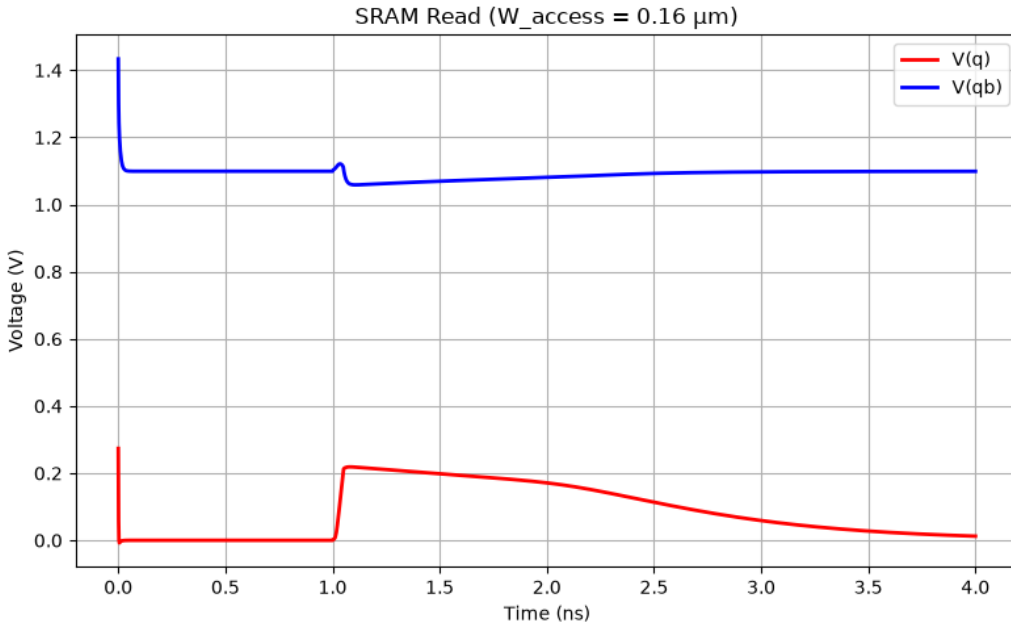


Figure 4: SRAM read transient at 85°C.

B CACTI: Size the 2 MB SRAM L2

Simulation Setup

CACTI 7 was used to evaluate the performance, energy, leakage, and area of the 2 MB SRAM cache at 45 nm. The baseline configuration uses a 64-byte cache block, 8-way associativity, four UCA banks, one read/write port, and an operating temperature of 360 K.

The cache size is

$$2 \text{ MB} = 2097152 \text{ bytes},$$

with a technology node of

$$45 \text{ nm}.$$

The cache organization and optimization settings are kept fixed for the baseline run. CACTI searches over the internal array organization and selects the configuration that minimizes the specified design objective.

B.1 Baseline 2 MB SRAM

For the baseline 2 MB configuration,

Parameter	Value
Cache capacity	2 MB
Technology	45 nm
Block size	64 B
Associativity	8-way
Number of banks	4
Read/write ports	1
Temperature	360 K
Access time	2.90184 ns
Dynamic read energy	0.791312 nJ
Leakage power	651.066 mW
Area	11.4742 mm ²

The winning data-array organization for this configuration is

$$(N_{dwl}, N_{dbl}, N_{spd}) = (4, 2, 1).$$

This organization is the configuration selected by CACTI for the baseline objective, where all design objective were given equal weight, and ED² optimization was used.

B.2 Cache Scaling with Capacity

To study the effect of cache capacity, the cache size was swept from 256 KB to 16 MB, while keeping the remaining configuration parameters fixed.

Capacity	$\log_2(\text{Capacity})$	Access Time (ns)	Area (mm ²)
256 KB	18	2.41714	4.96464
512 KB	19	2.48590	5.94616
1 MB	20	2.63524	7.59985
2 MB	21	2.90184	11.4742
4 MB	22	3.53084	18.4579
8 MB	23	4.46866	39.4935
16 MB	24	6.36235	74.6631

The access time increases monotonically with cache capacity. The increase is relatively small for the smaller capacities, but becomes progressively larger as the cache grows.

The corresponding access-time increase for each doubling is

Capacity doubling	Δt_{access} (ns)
256 KB $\rightarrow$ 512 KB	0.06876
512 KB $\rightarrow$ 1 MB	0.14934
1 MB $\rightarrow$ 2 MB	0.26660
2 MB $\rightarrow$ 4 MB	0.62900
4 MB $\rightarrow$ 8 MB	0.93782
8 MB $\rightarrow$ 16 MB	1.89369

Thus, the result does not show a constant one-gate-delay increase for every doubling. Instead, the incremental delay becomes significantly larger as the cache grows. This is particularly clear beyond 2 MB.

The increasing delay is a consequence of the larger physical array and the increased interconnect distance that signals must traverse. As the cache becomes larger, decoder, wordline, bitline, and hierarchical wire delays become increasingly important. Therefore, doubling capacity does not simply double the number of cells; it also changes the physical organization and the distance over which signals must propagate.

The area shows a much stronger scaling trend than the access time. The area increases from

$$4.96464 \text{ mm}^2$$

at 256 KB to

$$74.6631 \text{ mm}^2$$

at 16 MB.

Thus, a $64\times$ increase in capacity results in approximately a

$$\frac{74.6631}{4.96464} \approx 15.0\times$$

increase in area.

The dynamic read energy also increases with capacity, from

$$0.658021 \text{ nJ}$$

at 256 KB to

$$1.72648 \text{ nJ}$$

at 16 MB. This occurs because accessing a larger cache requires switching larger and more distributed portions of the array and its associated peripheral circuitry.

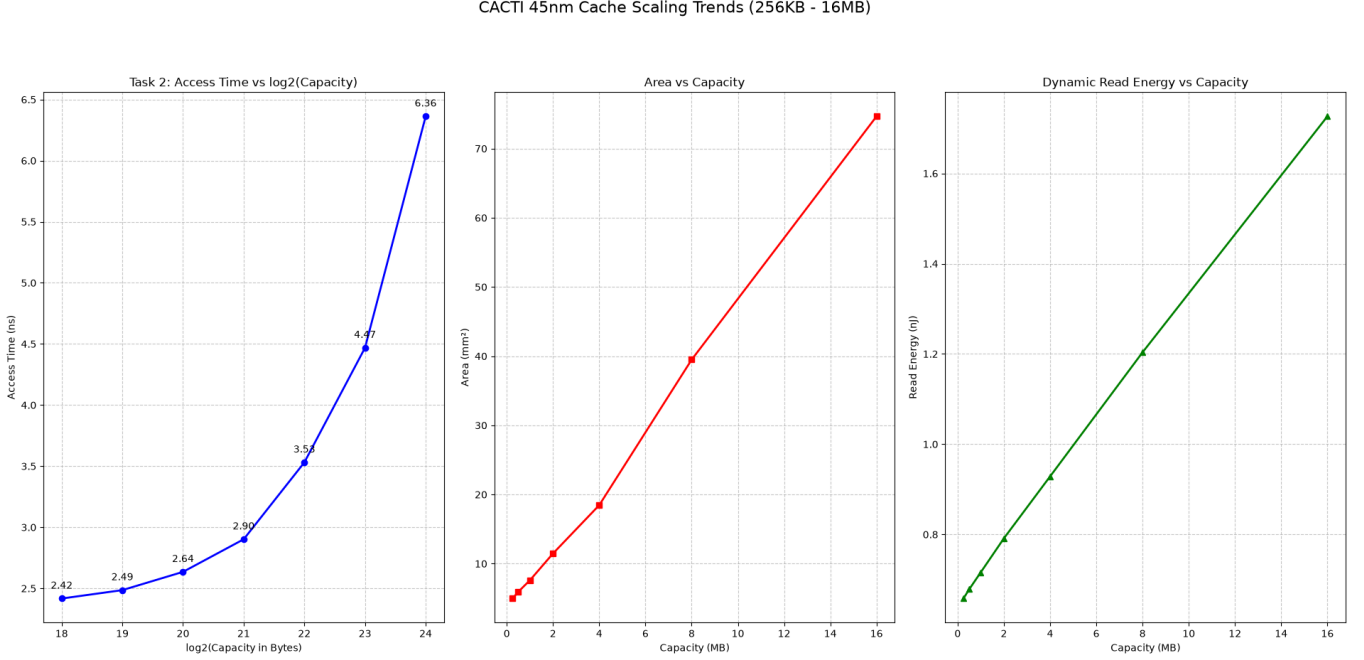


Figure 5: CACTI scaling results from 256 KB to 16 MB. The left plot shows access time versus $\log_2$ of capacity, while the middle and right plots show area and dynamic read energy, respectively.

The left plot in Fig. 5 directly answers the access time scaling question. The increasing slope at larger capacities shows that the one-gate-delay-per-doubling behavior is not a constant relationship for this cache organization.

B.3 Effect of the Design Objective

CACTI does not have a single universally optimal cache organization. The winning organization depends on the objective being optimized. To demonstrate this, the same 2 MB cache was optimized independently for pure delay, pure area, and ED^2 .

Configuration	N_{dwl}	N_{dbl}	N_{spd}	Access Time (ns)	Read Energy (nJ)	Area (mm ²)
Initial	4	2	1	2.90184	0.791312	11.4742
Pure delay	4	2	1	2.89231	0.898446	10.9695
Pure area	2	2	1	3.35518	0.796673	9.2449
ED^2	4	2	1	2.90184	0.791312	11.4742

For the pure-delay objective,

$$(N_{dwl}, N_{dbl}, N_{spd}) = (4, 2, 1)$$

is selected. This organization produces the lowest access time of

$$2.89231 \text{ ns.}$$

For pure-area optimization, CACTI instead selects

$$(N_{dwl}, N_{dbl}, N_{spd}) = (2, 2, 1)$$

with an access time of

$$3.35518 \text{ ns.}$$

The area-optimized configuration sacrifices timing in order to reduce the physical size of the array. For the ED^2 objective, the winning organization returns to

$$(N_{dwl}, N_{dbl}, N_{spd}) = (4, 2, 1)$$

with an access time of

$$2.90184 \text{ ns.}$$

The difference between the configurations can be understood from the meaning of the objective functions. A delay-optimized design is willing to use additional area or energy if this reduces the critical access path. An area-optimized design instead favors a more compact organization even if the resulting wires and subarrays produce greater delay. The ED^2 objective balances energy and delay, producing an organization close to the delay-optimized solution in this case.

The numerical results also show the trade-off directly. The delay-optimized configuration has an access time of

$$2.89231 \text{ ns,}$$

whereas the area-optimized configuration requires

$$3.35518 \text{ ns.}$$

Therefore,

$$\frac{3.35518 - 2.89231}{2.89231} \times 100 \approx 16.0\%$$

more access time is accepted by the area-optimized configuration.

This demonstrates that there is no unique “optimal” SRAM cache. The optimal organization depends on what the design is trying to minimize.

B.4 Bitline Delay

For the baseline 2 MB configuration, CACTI reports a data-array bitline delay of

$$t_{\text{BL,CACTI}} = 0.406901 \text{ ns.}$$

The corresponding data-array organization is

$$(N_{dwl}, N_{dbl}, N_{spd}) = (4, 2, 1),$$

and CACTI reports a subarray height of

$$0.336384 \text{ mm.}$$

The bitline length is therefore taken as

$$L = 0.336384 \text{ mm} = 336.384 \text{ } \mu\text{m}.$$

For the 45 nm technology, the standard semi-global interconnect parameters are approximately

$$R = 0.74 \text{ } \Omega/\mu\text{m}, \quad C = 0.291 \text{ fF}/\mu\text{m}.$$

Using the distributed RC approximation from the lecture,

$$t_{\text{BL,RC}} = 0.38RCL^2,$$

gives

$$t_{\text{BL,RC}} = 0.38(0.74)(0.291 \times 10^{-15})(336.384)^2,$$

and hence

$$\boxed{t_{\text{BL,RC}} \approx 9.26 \text{ ps} = 0.00926 \text{ ns.}}$$

The discrepancy between the simple RC estimate and CACTI is therefore

$$\frac{0.406901 - 0.00926}{0.406901} \times 100 \approx 97.7\%.$$

Equivalently, CACTI predicts a bitline delay approximately

$$\frac{0.406901}{0.00926} \approx 43.9\times$$

larger than the simple distributed-RC estimate.

This large discrepancy occurs because the $0.38RCL^2$ expression represents only an idealized distributed RC delay. CACTI models the complete SRAM subarray and its peripheral circuitry, including transistor loading, bitline drivers, sensing circuitry, and the detailed wire model. In addition, the actual CACTI interconnect model includes technology-dependent wire parameters and circuit effects that are not captured by a single uniform R , C , and L calculation.

Therefore, the hand calculation provides the expected first-order RC scaling, while the CACTI value represents the more complete cache-level timing model.

C NVSim: The Same 2 MB Cache Using STT-MRAM

Simulation Setup

NVSim was used to evaluate the same 2 MB cache using the supplied STT-MRAM cell model. The baseline MRAM cell has an area of $54F^2$ with aspect ratio 2.0 and

$$R_{\text{on}} = 3 \text{ k}\Omega, \quad R_{\text{off}} = 6 \text{ k}\Omega.$$

The read operation uses current sensing with

$$I_{\text{read}} = 40 \text{ }\mu\text{A},$$

while the baseline write currents are

$$I_{\text{reset}} = I_{\text{set}} = 200 \text{ }\mu\text{A}.$$

Both reset and set pulses have a duration of 10 ns.

The cache capacity is 2 MB with a 64 B cache line and 8-way associativity. The same 8-way associativity is used for the SRAM baseline from Part B so that the comparison is consistent.

C.1 Baseline STT-MRAM Results

Parameter	STT-MRAM
Cell area	$54F^2$
R_{on}	3 k Ω
R_{off}	6 k Ω
Read current	40 μA
Reset current	200 μA
Set current	200 μA
Cache capacity	2 MB
Cache line size	64 B
Associativity	8-way
Read latency	3.674 ns
Write latency	10.509 ns
Read energy	0.572 nJ/access
Write energy	1.538 nJ/access
Leakage power	437.413 mW
Area	0.682 mm ²

C.2 Comparison with the SRAM Baseline

The baseline STT-MRAM results are compared with the 2 MB, 8-way SRAM cache obtained from CACTI in Part B.

Metric	SRAM	STT-MRAM	Better
Read latency	2.902 ns	3.674 ns	SRAM
Write latency	Not reported	10.509 ns	–
Read energy	0.791312 nJ	0.572 nJ	STT-MRAM
Write energy	0.851627 nJ	1.538 nJ	SRAM
Leakage power	651.066 mW	437.413 mW	STT-MRAM
Area	11.4742 mm ²	0.682 mm ²	STT-MRAM

STT-MRAM provides clear advantages in **read energy, leakage power, and area**. Its read energy is approximately 27.7% lower than SRAM, while its leakage power is approximately 32.8% lower. The largest improvement is in area: STT-MRAM requires only 0.682 mm², compared with 11.4742 mm² for SRAM, corresponding to an area reduction of approximately 94.1%. This advantage comes primarily from the compact $54F^2$ STT-MRAM cell compared with the larger transistor-based SRAM cell. The lower leakage follows from the non-volatile nature of STT-MRAM.

SRAM provides better **read latency and write energy**. The read latency is 2.902 ns for SRAM compared with 3.674 ns for STT-MRAM. The write energy is 0.851627 nJ for SRAM compared with 1.538 nJ for STT-MRAM. The higher write energy of STT-MRAM results from the current-driven switching of the magnetic state of the MTJ.

A direct comparison of write latency is not made because the CACTI result used for the SRAM baseline does not report a corresponding write-latency value. Therefore, only the metrics for which both technologies have directly comparable reported results are used to declare one technology better.

Metric	Better	Reason
Read latency	SRAM	Faster electrical read operation
Read energy	STT-MRAM	Lower energy per read
Write energy	SRAM	No magnetic switching current
Leakage power	STT-MRAM	Non-volatile data retention
Area	STT-MRAM	Much smaller memory cell

Thus, STT-MRAM is strongly favored when **area, leakage power, and read energy** are important, whereas SRAM is favored for **read latency and write energy**. The comparison therefore highlights the fundamental trade-off between the high density and non-volatility of STT-MRAM and the higher speed and lower write cost of SRAM.

C.3 Effect of Increasing the TMR

The baseline cell has a resistance ratio

$$\frac{R_{\text{off}}}{R_{\text{on}}} = \frac{6}{3} = 2.$$

To study the effect of a higher TMR, the resistances were changed to

$$R_{\text{on}} = 4 \text{ k}\Omega, \quad R_{\text{off}} = 12 \text{ k}\Omega,$$

giving a resistance ratio of 3. The other parameters, including the 8-way associativity, were kept unchanged.

Parameter	Baseline	Higher TMR	Change
R_{on}	3 k Ω	4 k Ω	–
R_{off}	6 k Ω	12 k Ω	–
Read latency	3.674 ns	3.693 ns	+0.52%
Write latency	10.509 ns	10.517 ns	+0.08%
Read energy	0.572 nJ	0.573 nJ	+0.17%
Write energy	1.538 nJ	1.539 nJ	+0.07%
Leakage power	437.413 mW	437.413 mW	0.00%
Area	0.682 mm ²	0.682 mm ²	0.00%

The metric that moved the most was the **read latency**, which increased by approximately 0.52%. Increasing the TMR improves the resistance contrast between the two stored states, which should make the states easier to distinguish during sensing. However, the cache level improvement is small because the total read latency contains substantial peripheral and sense-amplifier delay. Therefore, a larger device-level TMR does not directly translate into a proportionally larger cache-level performance improvement.

C.4 Effect of Reducing the Write Current

The effect of reducing the write current was evaluated in two separate experiments. First, only the reset current was reduced from 200 μA to 100 μA , while keeping the set current at 200 μA . Second, both the reset and set currents were reduced to 100 μA .

Parameter	Baseline	Reset only	Reset + Set
I_{reset}	200 μA	100 μA	100 μA
I_{set}	200 μA	200 μA	100 μA
Read latency	3.674 ns	3.674 ns	3.674 ns
Write latency	10.509 ns	10.509 ns	10.494 ns
Read energy	0.572 nJ	0.572 nJ	0.571 nJ
Write energy	1.538 nJ	1.538 nJ	0.845 nJ
Leakage power	437.413 mW	437.413 mW	435.125 mW
Area	0.682 mm^2	0.682 mm^2	0.682 mm^2

Reducing only the reset current produces essentially no change in the cache-level results. In particular, the area, latency, energy, and leakage remain unchanged to the precision reported by NVSim. This is because the set current remains at 200 μA , so the larger write-current requirement is still present in the cell design.

When both currents are reduced to 100 μA , the most significant change is the **write energy**, which decreases from 1.538 nJ to 0.845 nJ, a reduction of approximately 45.1%. The write latency changes only slightly, from 10.509 ns to 10.494 ns, while the read energy and leakage power decrease only marginally. The reported area remains unchanged at 0.682 mm^2 .

The intended device-to-array coupling is

$$\text{write current} \rightarrow \text{access transistor size} \rightarrow \text{cell area} \rightarrow \text{array density}.$$

Therefore, reducing the required write current can potentially allow a smaller access transistor and hence a smaller memory cell. However, for the parameters used in this NVSim run, no reduction in the reported cache area was observed. Thus, the expected area benefit does not appear at the cache level for this particular configuration.

D Ramulator 2.0: DDR4 L2 Miss Stream

Simulation Setup

Ramulator 2.0 was used to study the memory traffic generated by the L2 cache. The workload is a repeated L2 miss trace containing both reads and writes across different ranks, banks, rows, and columns. The final trace contains 2048 accesses, consisting of 1536 reads and 512 writes.

The simulations use a DDR4-8Gb-x8 organization with two ranks and DDR4-2400 timing. The baseline configuration uses a single channel, open-row policy, RoBaRaCoCh address mapping, all-bank refresh, and the FRFCFS scheduler.

D.1 FRFCFS with One Channel

Statistic	Result
Memory cycles	768
Read requests	1536
Write requests	512
Total read latency	3916 cycles
Average read latency	81.5833 cycles
Total write latency	Not reported
Row-buffer hits	63
Row-buffer misses	1
Row-buffer conflicts	0
Row-buffer hit rate	98.4375%
Read throughput	4801.92 MB/s
Write throughput	1600.64 MB/s

The row-buffer hit rate is very high because most accesses remain within an already-open row. The simulator reports the total read latency directly, but the current controller statistics do not expose a corresponding write-latency counter. Therefore, no write-latency value is inferred from the available statistics.

D.2 Scheduler Comparison

The assignment asks for a comparison between FRFCFS and FCFS. However, the current Ramulator 2.0 build does not provide an FCFS scheduler; the available alternative used here is **FRFCFS-RowHit**. The comparison below therefore documents the available implementation rather than claiming a true FCFS result.

Statistic	FRFCFS	FRFCFS-RowHit
Memory cycles	768	768
Total read latency	3916 cycles	3916 cycles
Average read latency	81.5833 cycles	81.5833 cycles
Row-buffer hits	63	63
Row-buffer misses	1	1
Row-buffer hit rate	98.4375%	98.4375%

The measured latency difference is therefore 0 cycles. This should not be interpreted as an FCFS-versus-FRFCFS result because FCFS is not implemented in this build. Under a genuine FCFS policy, a row miss may be serviced before an older row hit, increasing the number of precharge and activate operations. The corresponding latency penalty would be determined primarily by t_{RP} , required to close the current row, and t_{RCD} , required to activate the next row before the column command can be issued.

D.3 Row Bits Below Bank Bits

For this experiment, the address mapping was changed so that bank selection occurs before the row field. This was implemented using the `PassThroughAddrMapper` with the address vector ordered as channel, rank, bank, row, and column.

Statistic	Baseline	Modified Mapping	Change
Memory cycles	768	768	0
Total read latency	3916	3203	-713
Average read latency	81.5833	118.6300	+37.0467
Row-buffer hits	63	2	-61
Row-buffer misses	1	8	+7
Row-buffer conflicts	0	29	+29
Row-buffer hit rate	98.4375%	20.0000%	-78.4375 percentage points
Total throughput	6402.56 MB/s	3601.44 MB/s	-2801.12 MB/s

The modified mapping substantially reduces the row-buffer hit rate from 98.44% to 20.00%. The number of row-buffer conflicts also increases from 0 to 29. Consequently, the average read latency rises from 81.58 to 118.63 cycles, while the total throughput falls by approximately 43.75%.

This occurs because the modified mapping distributes accesses across different rows, reducing the opportunity to reuse an already-open row. More row conflicts therefore require additional precharge and activate operations, reducing the effective bank-level parallelism.

The lower value of the total `read_latency` counter in the modified run should be interpreted carefully. The current frontend forwards or coalesces requests differently under this mapping, whereas the average per-read latency and throughput directly show the performance degradation caused by the loss of row locality.

D.4 Increasing the Channel Count

The final experiment uses two independent DDR4 channels with `CacheLineInterleave`. However, every address in the supplied trace has its channel field set to zero. Therefore, all requests are still directed to channel 0 while channel 1 remains idle.

Statistic	One Channel	Two Channels	Change
Memory cycles	768	768	0
Total read latency	3916	3916	0
Average read latency	81.5833	81.5833	0
Total throughput	6402.56 MB/s	6402.56 MB/s	0
Active row hits	63	63	0
Active row misses	1	1	0

Adding a second channel provides no measurable improvement because the workload does not exercise it. All requests continue to use channel 0, so the second controller remains idle.

For a workload that actually distributes requests across both channels, the binding timing constraint would be identified by examining the command sequence and timing gaps in the controller output. The main timing parameters are t_{RCD} , t_{RP} , t_{RAS} , and t_{CCD} . The first two dominate row activation and precharge overhead, t_{RAS} constrains the minimum active-row time, and t_{CCD} limits successive column commands. The current statistics do not expose a detailed per-command timing breakdown, so a more specific bottleneck cannot be identified from this run alone.

E gem5: Running the Complete System

Simulation Setup

gem5 was used to evaluate the system-level effect of the SRAM and STT-MRAM L2 cache configurations. The SRAM baseline uses a 2 MiB L2 with a 7-cycle hit latency, while the STT-MRAM configuration uses an 8 MiB L2 with a 14-cycle hit latency. Both configurations use an O3CPU, with DDR4-2400 as the backing memory system and a 4 GiB memory space.

The simulations were performed on the `bfs` and `sssp` kernels. In the current environment, the GAPBS executables were not available, so compact deterministic stand-in programs with the same kernel names were used.

Configuration	L2 Capacity	L2 Hit Latency	CPU
SRAM baseline	2 MiB	7 cycles	O3CPU
STT-MRAM alternative	8 MiB	14 cycles	O3CPU

E.1 O3CPU Comparison

Kernel	Cache	IPC	L2 Miss Rate	simSeconds
bfs	SRAM	0.681882	0.867524	0.000182
bfs	STT-MRAM	0.644431	0.867786	0.000192
sssp	SRAM	1.43357	0.876168	0.000285
sssp	STT-MRAM	1.38557	0.875978	0.000294

For both kernels, the SRAM configuration achieves higher IPC and lower simulated execution time. For `bfs`, the STT-MRAM configuration increases simulated time from 0.000182 to 0.000192 s, while for `sssp` it increases from 0.000285 to 0.000294 s.

The L2 miss rates are almost unchanged between the two cache configurations. Thus, in these experiments the larger STT-MRAM cache does not remove enough additional misses to compensate for its additional seven cycles of hit latency.

E.2 Does the Larger Cache Win?

The relative performance of the STT-MRAM configuration is summarized below.

Kernel	MRAM/SRAM IPC	MRAM/SRAM simSeconds	Winner
bfs	0.945077	1.05495	SRAM
sssp	0.966518	1.03158	SRAM

The deciding property is the **locality of the workload**. A larger cache is beneficial only when the additional capacity eliminates enough expensive misses to compensate for the increased latency of every L2 hit. For these workloads, the reduction in miss rate is too small, so the additional seven cycles of hit latency dominate and SRAM remains faster.

Therefore, the larger STT-MRAM cache does not win for either of the measured kernels under the simulated conditions.

E.3 TimingSimpleCPU

The winning configuration from Task 2 was rerun using TimingSimpleCPU to study the effect of out-of-order execution.

Kernel	CPU	Cache	IPC	L2 Miss Rate	simSeconds
bfs	TimingSimpleCPU	2 MiB SRAM	0.240794	0.880444	0.000515
sssp	TimingSimpleCPU	2 MiB SRAM	0.300071	0.887759	0.00136

The substantial reduction in IPC compared with the O3CPU results shows the effect of latency hiding through out-of-order execution. An O3CPU can continue executing independent instructions while an older cache miss or memory operation is outstanding. TimingSimpleCPU issues instructions in order, so much more of the cache and memory latency is directly exposed to the processor.

The L2 miss rate is primarily determined by the workload and cache capacity, whereas IPC and simulated execution time reflect how effectively the processor can hide the resulting latency.

E.4 System-Level Design Conclusion

The results do not support the unconditional conclusion that STT-MRAM is faster than SRAM. Part B showed that SRAM provides a lower L2 hit latency, while Part C showed that STT-MRAM offers a substantial density advantage at the cost of higher access latency. Part D provides the DDR4 memory-system context for the misses generated by the cache. Finally, the gem5 results show that the larger STT-MRAM cache does not outperform SRAM for either **bfs** or **sssp** in the present experiments. For these workloads, the additional cache capacity does not reduce the miss rate enough to compensate for the increased L2 hit latency. Hence, under the simulated conditions, the **2 MiB SRAM configuration is preferable**.

The main assumptions that should be examined before drawing a silicon-level conclusion are the workload, cache timing model, and memory-system configuration. First, the compact stand-in programs should be replaced by the actual GAPBS kernels and production graph inputs, since locality and working-set size determine whether the larger cache is useful. Second, the 7-cycle SRAM and 14-cycle STT-MRAM latencies should be validated against the detailed CACTI and NVSim results, including write behavior and other circuit-level effects. Third, the DDR4 organization, scheduler, address mapping, and channel configuration should be varied because the memory-system model affects the penalty associated with an L2 miss. These checks would determine whether the observed SRAM advantage persists across realistic workloads and system configurations.